

THE BASE- β BRANCH OF $3\alpha + 2\beta = \pi$: BOUNDARY WALKS, AND THE STRUCTURE OF A HYPOTHETICAL PRIME TILING

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ABSTRACT. This is a companion note to *A signed-direction invariant for triangle tilings, and the exclusion of primes $\equiv 3 \pmod{4}$* . That paper excludes every prime $N \equiv 3 \pmod{4}$ exceeding 3 except the base- β candidates $N = 3f^2 - e^2$, which are exactly the primes $\equiv 11 \pmod{12}$. Here we record what is known about the internal structure of a hypothetical tiling in that one open branch: the boundary walk equations and their solutions, the apex-mismatch configuration, and the alignment and far-region theorems. None of this is needed for the main theorem; it is collected here so that the main paper can state the exception cleanly and so that the structural facts are available to anyone attacking the branch.

Two results are worth isolating. First, at $m = 1$ the walk equations are completely solved in a regime: if $f^2 > 2e^2$ then each equal side of the target carries no b -edge, and if $f^2 > 2ef + e^2$ then the base carries exactly e of them, so $R_{\text{base}} \in \{e, 2e\}$. Second, for $e = 1$ this pins the base walk to a single family, reducing that entire subfamily — the candidates $N = 3f^2 - 1$, among them 11, 47, 107, 191, 431, 587, 971 below 1000 — to the question of whether one explicit walk is realizable.

We also record, as prominently as the positive results, which attacks on this branch are now known to be dead: the whole class of linear direction invariants, the metric bounds derived from them, and the hope of closing $e = 1$ by proving that every side carries at least two c -edges.

1. SCOPE AND CONVENTIONS

Throughout, the tile is the primitive $3\alpha + 2\beta = \pi$ triangle

$$(a, b, c) = (ef, f^2 - e^2, f^2), \quad \gcd(e, f) = 1, \quad 1 \leq e < f,$$

with a opposite α , b opposite β , c opposite γ , and $2\gamma = \pi + \alpha$. The base- β target is the isosceles triangle with base angles β ,

$$\text{equal sides } S = f^3 m, \quad \text{base } L = em(3f^2 - e^2), \quad N = m^2(3f^2 - e^2),$$

and N is prime exactly when $m = 1$. We write $D = 4f^2 - e^2$, so $\sin \alpha = e\sqrt{D}/(2f^2)$ and the tiling's coordinates lie in $\mathbb{Q}(\sqrt{D})$. Numbering of the form “Proposition 4.3” refers to the main paper [2]; results are stated here only when they are not there.

We use freely from [2]: the dissection basics and vertex figures, the corner figures (a base corner carries a single tile presenting β ; the apex carries exactly three tiles, each presenting α), the γ -trap (every side of the target carries at least one c -edge), the unsplittability of b , and the corner parallelogram.

2. BOUNDARY WALKS

A side of the target of length L is exactly partitioned by the tile edges lying on it, so it determines a *walk equation*

$$n_a ef + n_b (f^2 - e^2) + n_c f^2 = L, \quad n_a, n_b, n_c \geq 0,$$

together with $n_c \geq 1$ from the γ -trap. The results of this section solve those equations and then constrain the order in which the edges occur.

Theorem 1 (The base- β family at $e = 1$, $f = 2$). *The isosceles triangle with base angles β and sides $(8, 8, 11)$ admits no tiling by $N = 11$ copies of the tile $(2, 3, 4)$. Consequently the base- β branch is closed for $N = 11$, unconditionally.*

Proof. Write $\omega = \alpha/2$. The law of cosines gives $\cos \alpha = (2f^2 - e^2)/(2f^2)$, hence $\sin \omega = e/(2f)$; as $1 \leq e < f$ places this strictly in $(0, \frac{1}{2})$, *Niven's theorem* forces α/π irrational — for every (e, f) , with no appeal to [1] (`tile_alpha_irrational`). Irrationality pins the vertex figures: a vertex $x\alpha + y\beta + z\gamma = k\pi$ forces $y + z = 2k$ and $2x + z = 3y$, so a π -vertex is $\{3\alpha, 2\beta\}$ or $\{\alpha, \beta, \gamma\}$, each base corner carries *exactly one* tile (its β -vertex) and the apex *exactly three* (`vertex_pi`, `vertex_beta_corner`, `vertex_apex`); in particular a π -vertex carries at most one γ , since $2\gamma = \pi + \alpha > \pi$. Each side of ABC is partitioned into whole tile edges (a supporting line meets a convex tile in a face).

Two citation-free constraints now cut the base. (a) *Corner parallelogram*: the unique β -corner tile T_B has its b -edge a chord $[P, Q]$, matched (by the mod- f edge arithmetic) by a single tile T' whose α, γ are swapped, so $T_B \cup T'$ is a parallelogram with sides a, c ; hence P, Q are $(1, 1, 1)$ -vertices whose third tile carries β , and the second base edge from each corner lies in $\{a, c\}$. So a b -edge can occupy only positions $3, \dots, k-2$ of a base of k edges. (b) γ -trap: no boundary junction carries γ on both sides, and only a c -edge leaves the γ -state, so every side carries at least one c -edge.

At $e = 1$, $m = 1$ the base $Y = 3f^2 - 1$ has $q_{\text{base}} \equiv 1 \pmod{f}$, and the only compositions are $B_0 = \{b, c, c\}$ ($k = 3$), $B_1 = (a^f, b, c)$ ($k = f + 2$) and $B_2 = (a^{2f}, b)$ ($k = 2f + 1$). B_2 dies by (b). B_0 dies by (a) for *every* f : its single b would need a position in $3, \dots, 1$. At $f = 2$, B_1 also dies by (a): $k = 4$ leaves positions $3, \dots, 2$. So no base survives, and $N = 11$ admits no tiling. $\square$

Theorem 2 (Boundary walks of the base- β target at $m = 1$). *Let $(a, b, c) = (ef, f^2 - e^2, f^2)$ be the primitive $3\alpha + 2\beta$ tile ($1 \leq e < f$, $\gcd(e, f) = 1$) and let $m = 1$, so that the target is $(f^3, f^3, e(3f^2 - e^2))$ and $N = 3f^2 - e^2$. Write a walk along a side as $P \cdot a + Q \cdot b + R \cdot c$ equal to that side's length. Then both walk sets are cut out by the same linear form $\langle e, f, 1 \rangle$, the base at level $2e$ and the equal side at level f :*

$$\begin{aligned} \text{base: } (P, Q, R) &= (je + fp, e + fj, R), & pe + jf + R &= 2e, & j &\in \mathbb{Z}_{\geq 0}, \\ \text{side: } (P, Q, R) &= (qe + fp, fq, R), & pe + qf + R &= f, & q &\in \mathbb{Z}_{\geq 0}, \end{aligned}$$

with $p \in \mathbb{Z}$ and $P, R \geq 0$. In the thin regime $f > 2e$ the levels force short lists:

- (i) (trichotomy) the base is exactly one of $\{b^e, c^{2e}\}$, $\{a^f, b^e, c^e\}$, $\{a^{2f}, b^e\}$; in particular it carries exactly e b -edges;
- (ii) (dichotomy) each equal side is $\{a^e, b^f\}$ or $\{a^{fp}, c^{f-pe}\}$ for some $p \geq 0$.

Feeding in the γ -trap ($R \geq 1$ on every side, Remark 21) removes the thinness hypothesis altogether. For every (e, f) at $m = 1$:

- (iii) no equal side carries a b -edge — each is $\{a^{fp}, c^{f-pe}\}$, $p \geq 0$; so all boundary b -edges lie on the base;
- (iv) the base's b -count $Q = e + fj$ obeys $j(f - e) \leq e - 1$.

In particular $e = 1$ forces $j = 0$ for every f — the base carries exactly one b -edge — and $f > 2e$ forces $j = 0$, recovering (i) minus its $R = 0$ member.

Proof. Base. Reducing $P \cdot ef + Q(f^2 - e^2) + Rf^2 = e(3f^2 - e^2) \pmod{f}$ gives $Qe^2 \equiv e^3$, so $f \mid Q - e$ by $\gcd(e, f) = 1$; as $0 \leq Q$ and $0 < e < f$, the least nonnegative residue is e , whence $Q = e + fj$ with $j \geq 0$. Substituting and cancelling one factor f leaves the *halved equation*

$$Pe + j(f^2 - e^2) + Rf = 2ef. \tag{1}$$

Reducing (1) mod f gives $e(P - je) \equiv 0$, so $f \mid P - je$, say $P = je + fp$, $p \in \mathbb{Z}$; substituting into (1) and cancelling f gives $pe + jf + R = 2e$. Conversely every such triple solves the original equation. For $f > 2e$: from (1), $j(f^2 - e^2) \leq 2ef$; if $j \geq 2$ then $f^2 - e^2 \leq ef$, i.e. $f^2 - ef - e^2 \leq 0$, whereas $(f - 2e)(f + e) = f^2 - ef - 2e^2 > 0$ gives $f^2 - ef - e^2 > e^2 > 0$ — contradiction. If $j = 1$ then $f \mid P - e$ forces $P \geq e$, and (1) gives $e^2 + (f^2 - e^2) \leq 2ef$, i.e. $f^2 \leq 2ef$, i.e. $f \leq 2e$ — contradiction. So $j = 0$, whence $f \mid P$, $P = fp$ with $p \geq 0$, and $pe + R = 2e$ forces $p \in \{0, 1, 2\}$ and $R = e(2 - p)$, giving the three walks.

Side. The same reduction on $P \cdot ef + Q(f^2 - e^2) + Rf^2 = f^3$ gives $Qe^2 \equiv 0$, so $f \mid Q$, $Q = fq$; cancelling f leaves $Pe + q(f^2 - e^2) + Rf = f^2$, whose reduction mod f gives $f \mid P - qe$, say $P = qe + fp$, and cancelling f again gives $pe + qf + R = f$. For $f > 2e$: $q \geq 2$ would give $2(f^2 - e^2) \leq f^2$, i.e. $f^2 \leq 2e^2 < f^2$, absurd. If $q = 1$ then $Pe + Rf = e^2$ and $f \mid P - e$ force $P = e$, $R = 0$; if $q = 0$ then $f \mid P$, $P = fp$ and $pe + R = f$.

(iii) and (iv) are one computation, and need no thinness. Suppose a side walk has $q \geq 1$. From $P = qe + fp \geq 0$ we get $fp \geq -qe > -qf$ (using $e < f$ and $q \geq 1$), so $p > -q$, i.e. $p \geq 1 - q$. The level equation then gives

$$R = f - pe - qf \leq f - (1 - q)e - qf = (1 - q)(f - e) \leq 0,$$

contradicting $R \geq 1$. Hence $q = 0$, which is (iii). Identically for the base with $P = je + fp \geq 0$: if $j \geq 1$ then $p \geq 1 - j$ and

$$R = 2e - pe - jf \leq 2e - (1 - j)e - jf = e - j(f - e),$$

so $R \geq 1$ forces $j(f - e) \leq e - 1$, which is (iv) (and it holds trivially at $j = 0$ since $e \geq 1$). For $e = 1$ this reads $j(f - 1) \leq 0$, so $j = 0$; for $f > 2e$ we get $je < j(f - e) \leq e - 1$, so $j = 0$. $\square$

Theorem 3 (Apex mismatch: the pierced corner). *Let a base- β target be tiled at any scale m such that both equal sides end (at the apex) with a c -edge — automatic at $m = 1$, where the sides are b -free. Then, writing T_2 for the middle of the three apex tiles:*

- (i) *the outer apex tiles carry c on the boundary and b inward, and T_2 carries b and c inward, so exactly one inner apex ray carries T_2 's c -edge against a neighbour's b -edge: a T -junction at distance b from the apex, whose vertex figure on the neighbour's side is $(1, 1, 1)$ (one α , one β , plus the neighbour's γ);*
- (ii) *the remaining segment of T_2 's c -edge has length $c - b = e^2$, and $n_a ef + n_b(f^2 - e^2) + n_c f^2 = e^2$ has no solution in nonnegative integers, for any $1 \leq e < f$ with $\gcd(e, f) = 1$;*
- (iii) *hence some far-side edge crosses the far endpoint of T_2 's c -edge straight through: every such tiling contains a **pierced β -corner** at distance f^2 along an inner apex ray, and the angles on T_2 's side of the piercing line beyond the corner form $\{\alpha, \gamma\}$ or $\{3\alpha, \beta\}$.*

Proof. (i) The apex is 3α (three tiles, each α there); the two edges of an α -corner are b and c . The boundary edges at the apex are the sides' last edges, c by hypothesis, so the outer tiles point b inward and T_2 has $\{b, c\}$ inward. Whichever inner ray pairs T_2 's c with an outer b , the b -edge ends at distance b : its far endpoint is the γ -corner of its tile (b joins α to γ), and the vertex figure on that side of the straight c -edge sums to π with a γ present, hence is $(1, 1, 1)$ by the vertex classification. (ii) Any solution has $n_a = n_c = 0$ since $ef, f^2 > e^2$; then $n_b(f^2 - e^2) = e^2$ gives $n_b f^2 = (n_b + 1)e^2$, and $\gcd(e, f) = 1$ forces $f^2 \mid n_b + 1$, so $n_b \geq f^2 - 1$ and $e^2 = n_b(f^2 - e^2) \geq f^2 - 1 \geq e^2$, with equality forcing $f^2 - e^2 = 1$ — impossible, as $f^2 - e^2 \geq 2e + 1 \geq 3$. (iii) The far side of the leftover segment is covered by consecutive whole tile edges starting at the T -junction; by (ii) no partial sum lands exactly on the segment's far endpoint, so some edge strictly crosses it. That endpoint is the β -corner of T_2 (c joins α to β). The angles on T_2 's side of the crossing line beyond the corner fill $\pi - \beta$; writing them as

$x\alpha + y\beta + z\gamma$ and clearing β, γ , irrationality of α/π forces $y + z = 1$, $2x + z = 3y + 3$, whose only solutions are $(3, 1, 0)$ and $(1, 0, 1)$. $\square$

Remark 4 (Validation on the 44-tiling). The theorem can be checked against the genuine 44-tiling ($m = 2$; its equal sides do end with c), where it holds exactly: the apex rays pair one $b \mid b$ and one mismatch $c \mid b$; the middle tile's c -endpoint $V = (10, 2\sqrt{15})$ is pierced by a straight b -edge, and the sector on the middle tile's side is $\beta + \{\alpha, \gamma\}$ — the first continuation type. Parts (ii) and the vertex figures in (i), (iii) are machine-checked (`apex_leftover_nonrepresentable`, `pierced_corner_types`, `BaseBetaWalks.lean`). The configuration is an interior constraint, uniform in (e, f) and unconditional at $m = 1$: every hypothetical thick-regime tiling, $N = 59$ included, must contain it.

Proposition 5 (Rung two: the pre-piercer chain). *In the configuration of Theorem 3, every full far-side edge strictly between the T-junction and the piercer lies inside $[b, f^2]$, hence has length $\leq e^2$; among the tile edges only $b = f^2 - e^2$ can satisfy this, and only in the super-thick regime $f^2 \leq 2e^2$. So the piercer starts at $(t+1)b$ with $tb < e^2$, where $t = 0$ outside super-thick; and if the piercer is itself a b -edge, its corner at its starting junction is exactly α (the $(1, 1, 1)$ figure offers $\{\alpha, \beta\}$, while a b -edge's endpoints are $\{\alpha, \gamma\}$). In particular for $N = 59$ ($(e, f) = (4, 5)$, super-thick: $25 \leq 32$) the piercer starts at distance 9 or 18, and for every non-super-thick member it starts at the T-junction itself.*

Theorem 6 (Alignment of the mismatch ray). *Let $m = 1$, $e \geq 2$, and let r be the mismatch inner apex ray of Theorem 3, of inner length $f \cdot b$ (Proposition 5, $b = f^2 - e^2$). Along r , the near side is T2's single unsplittable c -edge covering $[0, f^2]$, and the far side begins with T1's b -edge. Then:*

- (i) no common junction lies in $(0, f^2)$: for $1 \leq k \leq f - 1$, $k \cdot b$ is never $n_a e f + n_b b + n_c f^2$ with $n_c \geq 1$ (so every far-side junction is a genuine T-junction, the near c -edge passing straight through it);
- (ii) the far side is exactly b^f : the only representation of $f \cdot b$ as $n_a e f + n_b b + n_c f^2$ with $n_b \geq 1$ is $(0, f, 0)$;
- (iii) $V = f^2$ is pierced: $b \nmid f^2$, so no far-side junction $k \cdot b$ equals f^2 and a far b -edge straddles T2's β -corner.

All three are machine-checked (`far_near_disjoint`, `far_is_bpow`, `b_not_dvd_fsq`); (ii) alone uses γ -vertex induction (via the walk arithmetic) to fix the first far edge, the rest is pure number theory.

Proof. (i) If $kb = n_a e f + n_b b + n_c f^2$ then $(k - n_b)b = f(n_a e + n_c f)$, so $f \mid (k - n_b)e^2$ and $\gcd(e, f) = 1$ give $f \mid k - n_b$; writing $k - n_b = fs$, the equation reads $sb = n_a e + n_c f \geq f > 0$ (as $n_c \geq 1$), so $s \geq 1$ and $k \geq n_b + f \geq f$, against $k \leq f - 1$. (ii) $f \mid n_b$ by the same reduction, and $n_b \leq f$ by size, so $n_b \in \{0, f\}$; $n_b \geq 1$ forces $n_b = f$, and then $n_a e f + n_c f^2 = 0$ gives $n_a = n_c = 0$. (iii) $b \mid f^2$ would give $b \mid e^2$, hence $b \mid \gcd(f^2, e^2) = 1$, so $b = 1$, impossible as $b \geq 2e + 1 \geq 5$. The far junctions are the multiples kb , none equal to f^2 , so the b -edge containing f^2 straddles it. $\square$

Theorem 7 (The mismatch ray is completely determined). *Let $m = 1$, $e \geq 2$, and let the family be thick, $f \leq 2e$. Then along the mismatch inner apex ray, of inner length $f b$:*

$$\text{far side} = b^f \text{ (} f \text{ edges),} \quad \text{near side} = \{a^{f-e}, c^{f-e}\} \text{ (} 2(f-e) \text{ edges),}$$

and the near side is the unique possibility. In particular the near side carries equally many a - and c -edges, and no b -edge.

Proof. The far side is Theorem 6(ii). For the near side, $T2$'s c -edge covers $[0, f^2]$ and the remainder $f b - f^2 = f(b - f)$ is covered by whole edges, say n_a of length a , n_b of length b and n_c of length c . Reducing modulo f kills n_b as in Theorem 6(i), and cancelling one factor f leaves

$$n_a e + n_c f + f = b.$$

Now $(n_a, n_c) = (f - e, f - e - 1)$ solves it, since $(f - e)e + (f - e - 1)f = f^2 - e^2 - f = b - f$. For uniqueness, $\gcd(e, f) = 1$ and the equation give $f \mid n_a + e$, so $n_a + e = ft$; $t = 0$ is impossible and $t \geq 2$ would force $n_a e \geq (2f - e)e$, whence $2ef \leq f^2 - f$, i.e. $f \geq 2e + 1$, contradicting thickness. Hence $t = 1$, $n_a = f - e$, and then $n_c f = f(f - 1 - e)$ gives $n_c = f - e - 1$. Adding $T2$'s own c -edge produces the multiset $\{a^{f-e}, c^{f-e}\}$, whose total length is $(f - e)(ef + f^2) = f(f - e)(e + f) = f b$ as required. $\square$

Theorem 8 (Unique representation of kb). *For $1 \leq k \leq f - 1$ the only way to write kb as $n_a e f + n_b b + n_c f^2$ in nonnegative integers is b^k . (Machine-checked, `kb_unique_rep`; it generalizes Theorem 6(ii), which needed $n_b \geq 1$.)*

Theorem 9 (The angle $(\pi - \alpha)/2$). *A vertex figure filling $(\pi - \alpha)/2$ is $\alpha + \beta$, and nothing else. (Machine-checked, `halfpi_minus_alpha_unique`.)*

Theorem 10 (The b -run orientation lemma). *Let a run of b -edges be laid end to end along a ray, seeded at an apex tile. Then every b -edge of the run carries α at its apex end and γ at its far end, and every junction interior to the run is an (α, β, γ) π -vertex. In particular the $3\alpha + 2\beta$ figure never occurs along such a run.*

Proof. A b -edge joins its tile's α - and γ -corners, so at a junction the incoming tile presents α or γ , and likewise the outgoing one. The $3\alpha + 2\beta$ figure contains no γ , and two γ 's cannot share a side of a straight line since $2\gamma = \pi + \alpha > \pi$. Hence if the incoming corner is γ the figure is $(1, 1, 1)$, whose single α must be the outgoing corner, so the next edge again ends in γ . An apex tile has α at the apex, hence γ at the far end of its b -edge, which seeds the induction; it proceeds with no branching. $\square$

Theorem 11 (The far region is a scaled tile). *Suppose the mismatch apex ray is a complete wall, of inner length $f b$. Then it cuts off a triangle with angle α at the apex, β at the base corner and hence γ at its third vertex, and with sides $f c$, $f a$, $f b$ — that is, the tile scaled by f . It therefore has area f^2 times the tile's and is tiled by the standard f -subdivision into f^2 copies, whose ray side is exactly b^f with the junction chain of Theorem 10.*

Corollary 12 (The far side can never give a contradiction). *Under the hypothesis of Theorem 11 — that the mismatch apex ray is a complete wall — no obstruction to a base- β tiling can be derived from the far side of that ray, at any (e, f) : that side admits a genuine tiling. The hypothesis may not be dropped; if the ray is not a complete wall the far region is not a scaled copy of the tile and the conclusion does not follow.*

Remark 13 (Where the obstruction must instead live). Corollary 12 is a negative result and an important one: it explains why the pierced-corner analysis resisted three independent attempts. The configuration it pins down is not contradictory but *realizable*, so no amount of further chasing on that side can close the branch. The check is exact for every family — for $(e, f) = (4, 5)$ the far region has sides $(45, 100, 125) = 5 \cdot (9, 20, 25)$ and contains $25 = f^2$ of the 59 tiles — and it is confirmed on the genuine 44-tiling, whose mismatch ray has *no* straddling tile and whose far region contains $16 = (mf)^2$ tiles with sides $4 \cdot (c, a, b)$.

The live direction is the other ray. For every thick member with $m = 1$, $e \geq 2$, the *matched* inner apex ray is *not* a wall. Indeed if it were, the target would split into $f^2 + f^2 + (f^2 - e^2)$ tiles, the last a middle triangle M with apex α , legs $f b$ and base $e b$. Since kb has the unique

representation b^k for $1 \leq k \leq f - 1$ (Theorem 8), both edges of M at its far corner are b -edges; but the angle there is $(\pi - \alpha)/2$, whose only decomposition is $\alpha + \beta$ (Theorem 9), and a b -edge endpoint is an α - or γ -corner, so no tile present can supply the β . This is the precise mechanism behind the failure of the clean-ray corollary retracted in Remark 14, and it is confirmed on the 44-tiling, whose matched ray carries exactly three straddling tiles against the mismatch ray's zero.

Consequently the branch now reduces to a single question: *can the residual region — the target minus the far wall, containing $N - f^2$ tiles and crossed by a necessarily straddled matched ray — be tiled?* Every tool developed here (walk arithmetic, vertex figures, unsplittability) is wall-based and goes silent once a tile straddles. What is needed is a bound on where a tile may straddle a ray whose two ends are pinned by b -edges.

Remark 14 (Two corrections this forces). Theorem 10 is uniform in (e, f) and in m , and is confirmed on the genuine 44-tiling, where all four far b -edges of the mismatch ray and both sides of the clean ray read $\alpha \rightarrow \gamma$. It strictly generalizes Proposition 5, which established only the first step. But it also refutes two statements of an earlier version, which we withdraw.

First, the assertion that the *clean* apex ray is b^f against b^f . Its justification — “by the symmetry of the isosceles target both inner rays have the same inner length $f b$ ” — conflates two objects. The symmetry does apply to the geometric rays; it does not apply to the tiling's maximal edge *segment* along a ray, which may terminate early at a vertex where the edges turn away. In fact the clean ray's segment can never reach the base: if it did, both final b -edges would contribute γ at an interior point of the base, where the tiles above sum to π , contradicting $2\gamma > \pi$. On the 44-tiling the two segments have lengths 12 and 9 respectively, and the clean ray terminates at an interior $\beta + 3\gamma$ vertex — which is precisely the M_4 vertex the census forces.

Second, and more seriously, the same objection applies to the hypothesis under which Theorems 6 and 7 were stated: that the *mismatch* ray's maximal segment attains the full length $f b$. That too was justified by symmetry, and it too now requires a genuine proof. The arithmetic of those theorems is unaffected — given a chain of total length $f b$ containing a b -edge, the conclusions stand and remain machine-checked — but their geometric hypothesis must be stated explicitly. Establishing that the mismatch segment reaches the base is the single highest-value open step in this branch: it would restore Theorems 6 and 7 unconditionally for every thick member.

Remark 15 (Scope and consequences). Thickness is used exactly once, and sharply: the next solution would be $n_a = 2f - e$, which needs $f \geq 2e + 1$ — the thin regime. So in the regime that is open, the entire mismatch ray is now pinned on *both* sides, with no freedom beyond the order of the near-side edges. Explicitly: $N = 23$ has far 5^3 , near $\{6, 9\}$; $N = 59$ far 9^5 , near $\{20, 25\}$; $N = 66$ far 16^5 , near $\{15^2, 25^2\}$; $N = 83$ far 11^6 , near $\{30, 36\}$; $N = 167$ far 39^8 , near $\{40^3, 64^3\}$. The count of interior junctions is therefore also determined — $f - 1$ on the far side and $2(f - e) - 1$ on the near side, none shared (Theorem 6(i)) — so a hypothetical thick tiling carries exactly $3f - 2e - 2$ T-junctions along this ray, each a π -vertex. The arithmetic is machine-checked (`near_side_unique`, `BaseBetaWalks.lean`).

Remark 16 (Scope, and why $m = 1$ is the rigid case). Theorem 2(i) generalizes the $e = 1$ base analysis used for Theorem 1 to the infinite family $f > 2e - e = 1$, $f \geq 3$; $e = 2$, $f \geq 5$; $e = 4$, $f \geq 9$; ... — covering the primes $N = 47, 71, 107, 191, 227, 239, 359, \dots$, and its parts (iii)–(iv), which need no thinness, deliver the $e \geq 2$ analogue of the $e = 1$ structure theorem of Remark 10 for *every* (e, f) : the equal sides are b -free and the base's b -count is bounded by $j(f - e) \leq e - 1$. The step that fails for $m \geq 2$ is the very first one: there $Q \equiv em \pmod{f}$, so j may be *negative* and the level rises to $2em$. This is not an artifact. The genuine 44-tiling ($e=1, f=2, m=2$) has base walk *aaaaccca*, i.e. $(P, Q, R) = (5, 0, 3)$, which is the parameter $j = -1$, $p = 3$; and the

99-tiling ($e=1, f=2, m=3$) has base walk $aabbbbbbbcc$, i.e. $(2, 7, 2)$, the parameter $j = 2, p = 0$. Both satisfy $R \geq 1$, as the γ -trap requires.

Theorem 17 (Walk structure at $m = 1$). *Let $m = 1$, $\gcd(e, f) = 1$, $1 \leq e < f$, and consider a tiling of the base- β target.*

- (i) *If $f^2 > 2e^2$, then each equal side carries no b -edge. Its edges are a 's and c 's with $n_a = fk$ and $n_c = f - ke \geq 1$, and it ends with a c -edge at the apex.*
- (ii) *If $f^2 > 2ef + e^2$ — equivalently $f/e > 1 + \sqrt{2}$ — then the base carries exactly e b -edges, and $n_a = f\ell$, $n_c = e(2 - \ell)$ with $\ell \in \{0, 1\}$. In particular*

$$R_{\text{base}} \in \{e, 2e\}.$$

Proof. A side of length L is exactly partitioned by the tile edges on it, giving $n_a ef + n_b b + n_c f^2 = L$ with $n_a, n_b, n_c \geq 0$, and $n_c \geq 1$ by Proposition 21.

(i) Here $L = f^3$. Rearranging with $b = f^2 - e^2$ gives $n_b e^2 = f(n_a e + n_b f + n_c f - f^2)$, so $f \mid n_b e^2$ and hence $f \mid n_b$ by coprimality; write $n_b = ft$. If $t \geq 2$ the b -edges alone cost at least $2fb = 2f^3 - 2fe^2$, which together with $n_c f^2 \geq f^2$ exceeds f^3 unless $f^2 \leq 2e^2$. If $t = 1$, cancelling f leaves $n_a e + n_c f = e^2$, so $n_a \leq e$ and $f \mid e(e - n_a)$; coprimality with $0 \leq e - n_a \leq e < f$ forces $n_a = e$ and then $n_c = 0$, contradicting $n_c \geq 1$. So $t = 0$. With $n_b = 0$ the equation is $n_a e + n_c f = f^2$, whence $f \mid n_a$; writing $n_a = fk$ gives $n_c = f - ke$. Finally, the apex tile on that side presents α (Proposition 20), whose incident edges are b and c ; as the side has no b -edge, it ends with a c -edge.

(ii) Here $L = e(3f^2 - e^2)$, and the same rearrangement gives $(n_b - e)e^2 = f(n_a e + n_b f + n_c f - 3ef)$, so $f \mid n_b - e$; write $n_b = e + ft$. Negative t makes $n_b < 0$. For $t \geq 1$ the b -edges alone cost at least

$$(e + f)b = (f + e)^2(f - e) = e(3f^2 - e^2) + f(f^2 - 2ef - e^2),$$

which already exceeds L under the hypothesis, before the mandatory c -edge is placed. So $n_b = e$; subtracting and cancelling f leaves $n_a e + n_c f = 2ef$, whence $f \mid n_a$, and writing $n_a = f\ell$ gives $n_c = e(2 - \ell)$. Then $n_c \geq 1$ forces $\ell \leq 1$.

Machine-checked (`equal_side_no_b`, `equal_side_shape`, `base_b_count` in `lean/BaseBetaWalkArith.lean`), and cross-checked by enumerating all walk solutions for the 198 coprime pairs with $f \leq 25$: no discrepancies. $\square$

Remark 18. Both hypotheses are sharp: $n_b = 2f$ on an equal side is admissible exactly when $f^2 \leq 2e^2$, and $n_b = e + f$ on the base exactly when $f^2 \leq 2ef + e^2$, by the displayed identity. Of the 42 base- β prime candidates below 1000, 28 satisfy (i) — including the thick values 23, 131 and 167, since (i) needs only $f/e > \sqrt{2}$ — and 17 satisfy (ii). The theorem structures those cases; it does not by itself exclude any of them.

Theorem 19 (The $e = 1$ subfamily reduces to one walk). *Let $e = 1$, $m = 1$ and $f \geq 3$, so the target is $(f^3, f^3, 3f^2 - 1)$, the tile is $(f, f^2 - 1, f^2)$ and $N = 3f^2 - 1$. Both hypotheses of Theorem 17 hold ($f^2 > 2$ and $f^2 > 2f + 1$). In any tiling:*

- (i) *neither equal side carries a b -edge; each equal side begins and ends with a c -edge, and therefore carries at least two;*
- (ii) *the base carries exactly one b -edge and exactly one c -edge — so $R_{\text{base}} = 1$ — and its first and last edges are a -edges.*

Thus the base walk is a permutation of (a^f, b, c) beginning and ending with a , with the b in position $3, \dots, f$.

Proof. (i) An equal side has length f^3 and its edges lie in $\{f, f^2 - 1, f^2\}$. Reducing $n_a f + n_b(f^2 - 1) + n_c f^2 = f^3$ modulo f gives $f \mid n_b$. If $n_b \geq 2f$ then $n_b(f^2 - 1) \geq 2f^3 - 2f > f^3$ for

$f \geq 2$. If $n_b = f$ then $f(f^2 - 1) = f^3 - f$ leaves f , forcing $n_c = 0$ and $n_a = 1$, hence no c -edge on that side — contradicting Proposition 21. So $n_b = 0$, and then $n_a + n_c f = f^2$, so $f \mid n_a$; write $n_a = fk$, $n_c = f - k$.

By Proposition 20 the apex tile on that side presents α , whose incident edges are b and c ; since the side carries no b -edge, it ends with a c -edge. Its other end is handled in (ii): there we show the base begins and ends with a -edges, so the base-corner tile's c -edge lies on the equal side, which therefore begins with a c -edge. Were $n_c = 1$ the single c -edge would be both the first and the last, forcing the side to consist of one edge and $f^3 = f^2$; so $n_c \geq 2$.

(ii) The base has length $3f^2 - 1$. Reducing modulo f gives $n_b \equiv 1$. If $n_b \geq f + 1$ then $n_b(f^2 - 1) \geq f^3 + f^2 - f - 1 > 3f^2 - 1$ for $f \geq 3$, so $n_b = 1$. Subtracting, $2f^2 = n_a f + n_c f^2$, so $n_a = f(2 - n_c)$ with $n_c \in \{0, 1, 2\}$, and $n_c \geq 1$ by Proposition 21.

If $n_c = 2$ then $n_a = 0$ and the base is three edges, a permutation of (b, c, c) . By Proposition 24 the first two edges lie in $\{a, c\}$ and so do the last two; with no a -edge available this forces all three to be c -edges, contradicting $n_b = 1$. Hence $n_c = 1$ and $n_a = f$, so the base is a permutation of (a^f, b, c) with $n = f + 2$ edges, and Proposition 24 places the b away from the first two and last two positions.

It remains to exclude $E_1 = c$. Suppose it. Then the unique c -edge is E_1 , so every one of $E_2, \dots, E_n$ is an a - or b -edge and places a γ at one of its endpoints; there are $f + 1$ of them and $f + 1$ interior junctions $J_1, \dots, J_{n-1}$. The corner tile T_A has its c -edge on the base, so its α vertex is J_1 and its γ vertex lies on the equal side; by Proposition 24 its partner T_3 presents γ at J_1 . So J_1 already carries a γ , and since a π -vertex carries at most one (Proposition 19), E_2 must place its γ at J_2 ; inductively E_i places at J_i , and E_n places at J_n , a corner — contradicting Proposition 20. Hence E_1 is an a -edge, and the same argument at B gives E_n an a -edge. This is what (i) used. $\square$

Remark 20. Theorem 19 reduces the whole $e = 1$, $m = 1$ subfamily — which contains the base- β prime candidates $N = 3f^2 - 1$, among them 11, 47, 107, 191, 431, 587 and 971 below 1000 — to the single question of whether that one base walk is realizable. It also corrects a plausible-looking plan: one might hope to close the subfamily by proving $R \geq 2$ on every side. That is false for the base. The walk with $R_{\text{base}} = 2$ is exactly the one the corner parallelogram kills, and the survivor has $R_{\text{base}} = 1$; on the base, “ $R \geq 2$ ” is not a lemma reducing the problem but a restatement of it. On the two equal sides $R \geq 2$ does hold, and is proved above.

Remark 21 (The $e \geq 2$ residue). The $e \geq 2$ residue ($N = 23, 59, 71, 83, 131, 167, \dots$; the members 23 and 71 are now settled by search, Table 1) resists all three natural invariant methods, and in two cases *provably*:

- **Linear/colouring invariants** — dead by Remark 16.
- **b -edge census.** On a maximal segment of length L the two sides carry $q_1 \equiv q_2 \pmod{f}$ (both $\equiv -e^{-2}L$), and the entire census — in *every* modulus, and in its per-direction refinement — is a formal consequence of the single parity identity $N(a + b + c) \equiv 2X + Y \pmod{2}$, valid for all (e, f, m) . So no congruence of this type can bite.
- **Euler / vertex counting.** For any such tiling $V - E + F = 2$ is *identically* the sum of the three corner identities $\#\alpha = \#\beta = \#\gamma = N$; the vertex-type system collapses to $Q_{013} = 1 + P_{320} + 2Q_{640} + Q_{431} + W_{320}$ and $3P_{320} + P_{111} + 6Q_{640} + 4Q_{431} + 2Q_{222} + 3W_{320} + W_{111} = N - 3$, which is feasible for every $e \geq 2$, $m = 1$ target. So Euler carries no information here. Here W_{320} and W_{111} count the interior vertices at which some tile is met in the *relative interior* of one of its edges: such a tile contributes a straight angle, so the remaining tile angles sum to π rather than 2π and the figure is $(3, 2, 0)$ or $(1, 1, 1)$ (Proposition 19). These types are not optional — the dissections here are not

edge-to-edge (Remark 14) — and an earlier version of this census omitted them. Their presence only enlarges the feasible set, so the conclusion is unaffected.

What *is* gained is citation-free structure, valid for all (e, f, m) : the γ -trap (Proposition 21) and the corner parallelogram (Proposition 24), both proved above from the corner figures and the unsplittability of b , and both used by the exact search as prunes. They were checked against the two genuine certificates before being proved: the first-two-edges rule of Proposition 24 holds on all six sides of the 44- and 99-tilings, and at each of the four base corners exactly one of the two sides begins with a c -edge.

This isolates the residue to one clean statement. Since $q_{\text{base}} \equiv em \pmod{f}$ while $q_{\text{equal}} \equiv 0$, we have:

if the base carries no b -edge, then $f \mid em$, hence $f \mid m$, hence no tiling with $m = 1$ — for every e at once.

The 44-tiling ($m = 2$) indeed carries *zero* b -edges on its entire boundary, consistent with this. But Remark 17 now shows the converse direction is a dead end: the 99-tiling has $f \nmid m$ and its base carries seven b -edges, so “the base never carries a b -edge” is *false* in general, and the displayed implication — whose hypothesis is equivalent to $f \mid m$ — cannot be used to exclude $m = 1$ without already knowing $f \mid m$, which is false. A fourth method also fails, and provably: *boundary counting*. Each side S of ABC , cut into k_S whole edges, forces at least $2k_S - 1$ tiles to meet it (the k_S inner tiles, plus a distinct “middle” tile at each of the $k_S - 1$ junctions, since a triangle has no two collinear edges); this is exactly tight on the 44-tiling (base $k = 8$, $15 = 2 \cdot 8 - 1$ tiles). But at $m = 1$ the boundary is *minimal* — for $e = 1$ the least feasible total is $\sum k_S = 3f + 2$ — so counting sees only $O(f)$ of the $N = \Theta(f^2)$ tiles, and $\sum(2k_S - 1) = 6f + 1$ exceeds $N = 3f^2 - 1$ only at $f = 2$, the case already closed. The ratio $(6f + 1)/(3f^2 - 1) \rightarrow 0$: $m = 1$ is the case *least* constrained by counting, not the most.

So no *invariant* route survives: linear invariants, the census, Euler and counting are all provably vacuous, and the one arithmetic statement that would settle it ($f \mid m$) is now known to be false. What does survive is *structure*. Theorem 2 classifies the boundary walks for every (e, f) at $m = 1$, and in the thin regime $f > 2e$ reduces the whole branch to two possible bases and b -free equal sides — the $e \geq 2$ analogue of Remark 10, and the first statement in this branch that is uniform in e . Parts (iii)–(iv) need no thinness at all: at $m = 1$ *no equal side carries a b -edge*, for every (e, f) , so all boundary b -edges lie on the base; and the base’s b -count $Q = e + fj$ obeys $j(f - e) \leq e - 1$. It does not close the branch, but it does locate the difficulty exactly. The bound $j \leq (e - 1)/(f - e)$ is uniform, and it degrades precisely as f falls towards e : at $e = 1$ it pins $j = 0$ for every f (one b -edge on the base, the $e = 1$ rigidity of Remark 10, now a corollary); at $f > 2e$ it still pins $j = 0$; but in the *thick* regime $f \leq 2e$ it leaves $\lfloor (e - 1)/(f - e) \rfloor + 1$ values of j and the trichotomy dissolves. That is precisely where the open members sit — $N = 59$ is $(e, f) = (4, 5)$, so $j \leq 3$; $N = 66$ is $(3, 5)$, so $j \leq 1$; likewise $N = 23, 83, 131, 167$. We record the branch as open and settle its members one at a time by exhaustive search. Table 1 lists every member below 110; of the eleven, eight are settled and 59 and 66 are now excluded by exhaustive search (apex-down with the walk prune; 1 838 175 and 7 232 464 nodes), leaving 83 under search. ($N = 83$, $(e, f) = (5, 6)$, is the smallest base- β prime not yet settled.) The walk classification also feeds back into the search: the γ -trap and the corner parallelogram have the multiset corollaries $R \geq 1$ on every side, $Q_{\text{base}} \leq k - 4$ and $Q_{\text{side}} \leq k - 2$ (k the number of edges of the walk), and the engine now prunes any partial boundary walk that fits no surviving multiset (prune P5). The rule was validated positionally on all six sides of the two genuine tilings before use — the 44-tiling’s base *aaaaccca* and the 99-tiling’s base *aabbbbbbbcc*, which is exactly tight for $Q \leq k - 4$ — the strengthened engine re-finds the 44-tiling from scratch (852,093 nodes; the found tiling passes the independent verifier: congruence, exact area, containment, pairwise disjointness) — and the corner rule is applied only after checking that the tile’s b -edge

is unsplittable ($n_a a + n_c c = b$ has no solution), which is what forces the parallelogram. For $N = 59$ this cuts the admissible bases from seven to four, for $N = 66$ from four to three.

3. WHAT IS KNOWN TO BE DEAD

The following approaches to this branch are closed, and are recorded so that they are not retried.

- (1) **The whole class of linear direction invariants.** Weighting a directed edge by a function of its direction alone, antisymmetric under $\theta \mapsto \theta + \pi$, yields a one-parameter family of identities; every member is a specialisation of the master identity of [2, Rem. 4.9]. The associated ideal-membership condition is satisfied identically for every root of unity and every (e, f) , and the optimal ℓ^1 bound the family can produce is $m(3f - 2e)(f + e)/f$, which is smaller than N by a factor at least $11/6$ and growing like mf . So no choice of weight, kernel, moment ladder or duality can exclude a candidate. Both statements are machine-checked in `lean/LambdaFactor.lean`.
- (2) **Closing $e = 1$ by proving $R \geq 2$ on every side.** False at the base: the walk with $R_{\text{base}} = 2$ is exactly the one the corner parallelogram kills, and the survivor has $R_{\text{base}} = 1$. On the base, “ $R \geq 2$ ” restates the remaining problem rather than reducing it. It does hold on the two equal sides, and is proved there.
- (3) **Boundary tile counts.** The number of tiles meeting ∂ABC grows like the number of boundary edges, and on the genuine certificates it is 28 of 44 and 44 of 99 — a falling fraction. Comparing it against N therefore never yields a contradiction.
- (4) **The far side of the mismatch ray,** under the hypothesis that the ray is a complete wall: that side is the tile scaled by f , and is tileable. No obstruction can come from there.

REFERENCES

- [1] M. Laczkovich, *Tilings of triangles*, Discrete Math. 140 (1995).
- [2] V. Bonfioli, *A signed-direction invariant for triangle tilings, and the exclusion of primes $\equiv 3 \pmod{4}$* , 2026.
- [3] M. Beeson, *Triangle tiling III: the case $3\alpha + 2\beta = \pi$* , arXiv:1206.2231.
- [4] M. Beeson, *Tilings of an isosceles triangle*, arXiv:1206.1974.

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